Gayatri Vidya Parishad College of Engineering Autonomous

Data Science | Course Code: DS301

Applied Statistics & Data Science - Applied Statistical Inference & Hypothe

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Overview: Central limit theorem, Student t-test, Chi-square independence tests, confidence inte

I. UNIT-WISE SYLLABUS BREAKDOWN & HIGH-YIELD TOPICS

Unit 1: Probability Foundations & Sampling
Random variables, Expectation, Variance, Joint distributions, Central Limit Theorem, Standard normal

Unit 2: Parametric Hypothesis Testing
Null & Alternative hypotheses, Type I and Type II errors, p-value interpretation, One-sample & Two-s

Unit 3: Categorical Analysis & ANOVA
Chi-Square goodness of fit and contingency tables, One-way and Two-way Analysis of Variance ANOVA, F

Unit 4: Correlation & Multivariate Analysis
Pearson and Spearman rank correlation, Multicollinearity, VIF, Residual diagnostics, Homoscedasticit

II. HIGH-YIELD SEMESTER EXAMINATION QUESTIONS

1. Formulate a two-tailed Student t-test for comparing independent sample means with unequal variances We
2. Construct an ANOVA summary table for a single-factor experiment with 4 treatment groups and calculate
3. Explain the Central Limit Theorem and prove its convergence using moment generating functions.

III. PRESCRIBED TEXTBOOKS & ACADEMIC REPOSITORIES

* Douglas C. Montgomery, George C. Runger - Applied Statistics and Probability for Engineers Wiley
* Gareth James et al. - An Introduction to Statistical Learning ISLR Springer

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